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College of Computer
Information Technology Dept.



المملكة العربية السعودية
وزارة التعليم
جامعة القصيم
كلية الحاسب
قسم تقنية المعلومات

Traffic Congestion Detection Using Deep Learning

Students:

Abdualziz Alsulaim	391110124
Fares Alharbi	391108062
Jawad Alsallom	412115937
Faisal Almosaimer	392111887

Supervisor:

Dr. Suliman Aladhadh

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Abstract

Traffic congestion in urban areas is a growing concern, leading to increased travel time, pollution, and reduced economic productivity. The aim of this project is to develop an efficient system for real-time traffic congestion detection using a combination of YOLO (You Only Look Once) for vehicle detection, Deep SORT (Simple Online and Realtime Tracking) for vehicle tracking, and OpenCV for image processing. This integrated approach will enable accurate detection and tracking of vehicles in complex traffic scenarios, providing valuable insights into congestion patterns.

Keywords: Deep Learning (DL), Machine Learning (ML), DeepSort (DS), You look only once (YOLO), OpenCV,

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Chapter 1

INTRODUCTION

INTRODUCTION

1.1 Introduction

Traffic congestion has become a pressing issue in Saudi Arabia, especially in urban centers such as Riyadh, Jeddah and Dammam, where rapid urbanization and population growth have led to an increase in the number of vehicles on the roads. In Saudi Arabia, traffic congestion is one of the country's most complex challenges, significantly impacting daily life by consuming the average driver's time and precious energy. This problem is particularly prominent in Riyadh, a city the size of the United States, where the already strained infrastructure is expected to face even greater demand following a decision by His Highness the Crown Prince Mohammed bin Salman to double the city's population by 2030 [1]. Due to Riyadh's outdated urban design, upgrading the infrastructure poses significant and seemingly impossible challenges, making traffic congestion a major and growing concern.

With advancements in artificial intelligence (AI) and machine learning, it is now possible to predict and manage traffic congestion proactively, both before and during peak traffic periods. By analyzing data from Sensors, traffic cameras, and other sources which we will discuss in this research, AI-driven systems can predict congestion patterns and alert drivers to upcoming traffic in real-time. This research explores the application of AI techniques to traffic congestion prediction, offering innovative solutions tailored to the unique needs of Saudi cities. In doing so, it aims to improve urban mobility, providing a viable alternative to infrastructure changes while aligning with Saudi Arabia's Vision 2030 goals to enhance the quality of life in its urban areas.

1.2 Problem Specification and Motivation

The rapid population expansion in Saudi Arabia, and Riyadh in particular, driven by strategic growth initiatives such as Vision 2030, has placed increasing pressure on existing, dilapidated road networks. For example, Riyadh is expected to double its population by 2030, creating urgent transportation challenges that the city's infrastructure was not originally designed to manage. Unlike many major global cities with flexible urban layouts, Riyadh's older and more rigid infrastructure limits options for road widening or other traditional methods of easing congestion. This structural rigidity, coupled with rising vehicle numbers, has led to severe traffic congestion and delays, directly impacting residents quality of life by adding tens of hours per week to their commute, increasing fuel costs, and amplifying air pollution. There is therefore an urgent need for an advanced and adaptive approach to traffic management that can alleviate congestion under these constraints.

The motivation and the goal behind applying artificial intelligence (AI) for traffic congestion prediction lies in its capacity to process large-scale, real-time data from diverse sources and generate proactive, data-driven responses to traffic patterns. Through machine learning algorithms, AI can analyze complex datasets from sensors, GPS devices, and traffic cameras, identifying trends and generating early warnings for potential traffic build-ups. This predictive capability offers an innovative alternative to infrastructure-heavy solutions, providing an agile response mechanism that traditional systems lack. By enabling real-time adjustments to traffic flow and offering drivers timely rerouting options, an AI-driven approach has the potential to significantly reduce congestion impacts. Furthermore, such a solution aligns closely with Vision 2030's commitment to sustainability and enhanced urban living standards, contributing to a smarter, more efficient future for Saudi Arabia's cities.

1.3 Goals and Objectives

This research aims to advance the application of artificial intelligence in traffic congestion detection while contributing to the scientific understanding and improvement of deep learning-based traffic management systems. The key objectives include:

- **Enhancing Detection and Tracking Performance:** Focus on improving the F1 score of the YOLO and DeepSort-based model to achieve higher accuracy in vehicle detection and tracking, ensuring robust results even in challenging traffic scenarios.
- **Building on Existing Research:** Develop and refine existing methodologies in the field of traffic analysis by integrating innovative techniques for feature extraction, multi-object tracking, and speed flow estimation.
- **Optimizing Congestion Analysis:** Expand the model's capability to not only detect congestion but also analyze contributing factors, such as lane-specific delays and irregular vehicle movements, to provide actionable insights for traffic management systems.
- **Logging Congestion Events and Real-Time Alerts:** Design a system to accurately log congestion events and trigger real-time alerts, enabling proactive traffic management and timely decision-making for improved flow and safety.

These objectives illustrate the flexible and extensive potential of AI applications in traffic management, with countless possibilities for enhancing urban mobility. This research aspires to pioneer a comprehensive, adaptive AI solution that transforms traditional traffic systems into responsive, predictive networks that meet the growing demands of Saudi Arabia's urban centers.

1.4 Study Scope

These objectives aim to enhance AI-based traffic management by improving detection accuracy, advancing current research, optimizing congestion analysis, and enabling real-time alerts, contributing to safer and more efficient transportation systems.

1.5 Study Plan and Schedule

Figure 1.1 and Figure 1.2 show the Gantt charts we followed during the first and second phases of our project, respectively.

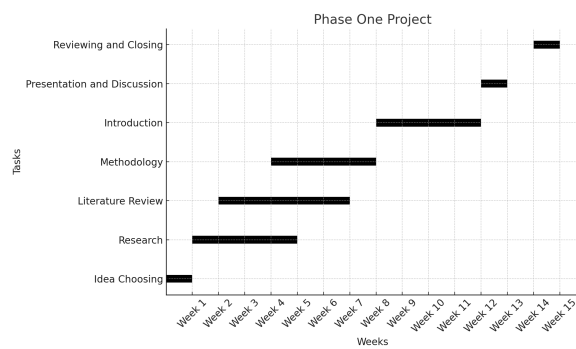


Figure 1.1: Gantt chart for Phase One

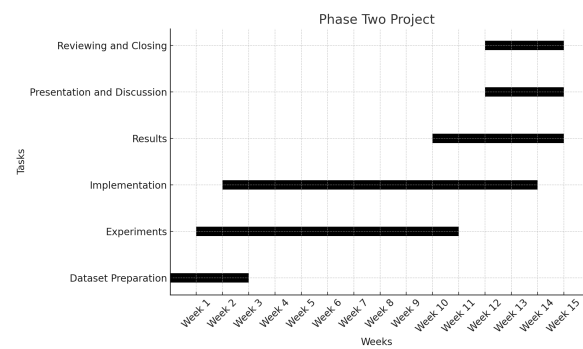


Figure 1.2: Gantt chart for Phase Two

1.6 Summary

In this chapter, we address the pressing issue of traffic congestion in Saudi Arabia's urban centers, especially Riyadh. Traditional infrastructure solutions are limited, so this research explores AI-driven methods for predicting and managing traffic, aiming to improve urban mobility and align with Vision 2030 for sustainable city living.

Chapter 2

LITERATURE REVIEW

LITERATURE REVIEW

2.1 Introduction

In this chapter, we will explore real-time traffic management tools, focusing on key concepts such as **machine learning (ML)**, **YOLO**, **OpenCV**, **DeepSort (DS)**, and semantic search. We will also review previous research in this field and examine the current challenges.

2.2 Background

In this section, we will provide clear definitions of DS, OpenCv , YOLO, DL, and discuss search

2.2.1 DeepSort

DeepSort tracking is a computer vision technique that involves automatically tracking the movement and position of an object in a video sequence over time. The goal of object tracking is to maintain the identity of the object, even as it moves and changes appearance, and to accurately locate it in each frame of the video [2]. it is a tracking algorithm that extends the capabilities of object detection algorithms by associating detected objects across frames and maintaining unique identities for each object. It leverages deep learning techniques to learn the appearance features of objects and uses them to track and match objects over time. [3]

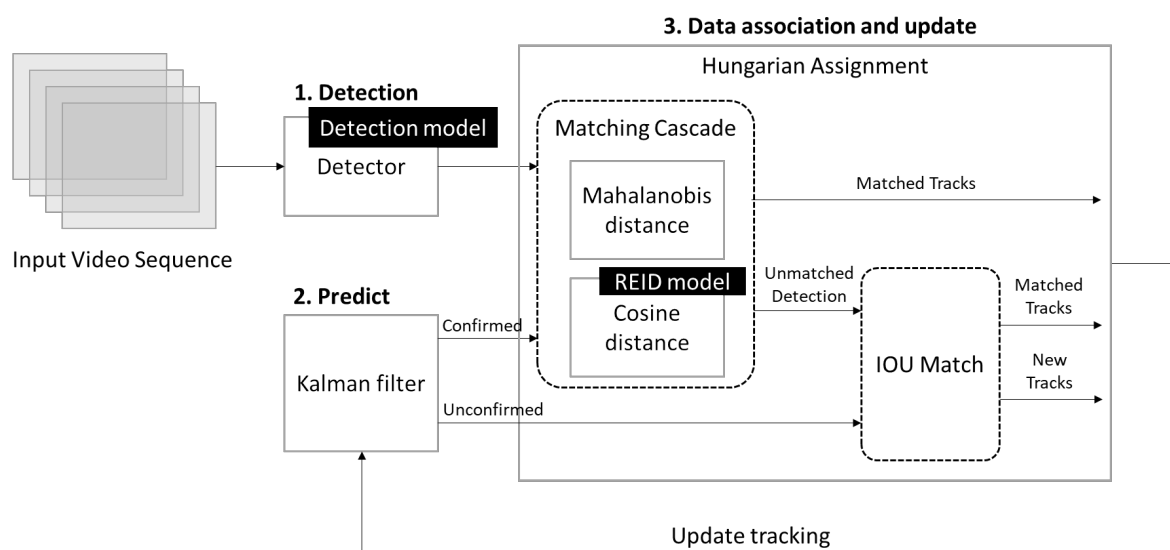


Figure 2.1: DeepSort Figure

2.2.2 Open CV

OpenCV (Open Source Computer Vision Library) is an open source computer vision and machine learning software library. OpenCV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in the commercial products. Being an Apache 2 licensed product, OpenCV makes it easy for businesses to utilize and modify the code. The library has more than 2500 optimized algorithms, which includes a comprehensive set of both classic and state-of-the-art computer vision and machine learning algorithms. These algorithms can be used to detect and recognize faces, identify objects, classify human actions in videos, track camera movements, track moving objects, extract 3D models of objects, produce 3D point clouds from stereo cameras, stitch images together to produce a high resolution image of an entire scene, find similar images from an image database, remove red eyes from images taken using flash, follow eye movements, recognize scenery and establish markers to overlay it with augmented reality, etc. OpenCV has more than 47 thousand people of user community and estimated number of downloads exceeding 18 million. The library is used extensively in companies, research groups and by governmental bodies.[4] .

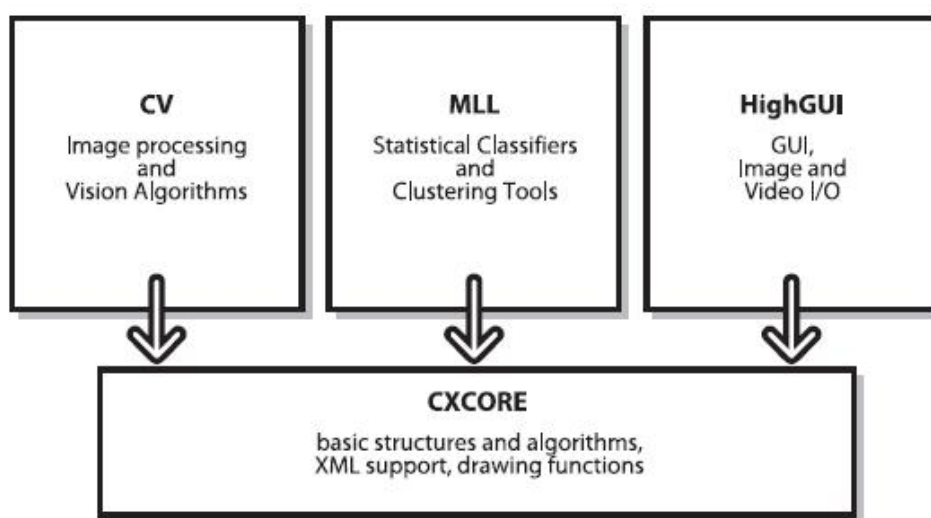


Figure 2.2: Open CV Figure

2.2.3 YOLO MODEL

YOLO stands for "**You Only Look Once**" model is The most effective object identification model was created by 58 open source contributors over the course of continuous development, and it is called YOLOv8, it is the state-of-the-art object detection algorithm Most contemporary object detection algorithms repurpose CNN classifiers with an aim of performing detections. For

instance, to perform object detection, these algorithms use a classifier for that object and test it at varied locations and scales in the test image. However, YOLO reframes object detection, i.e., instead of looking at a single image thousand times to perform detection, it just looks at the image once and performs accurate object predictions [3].

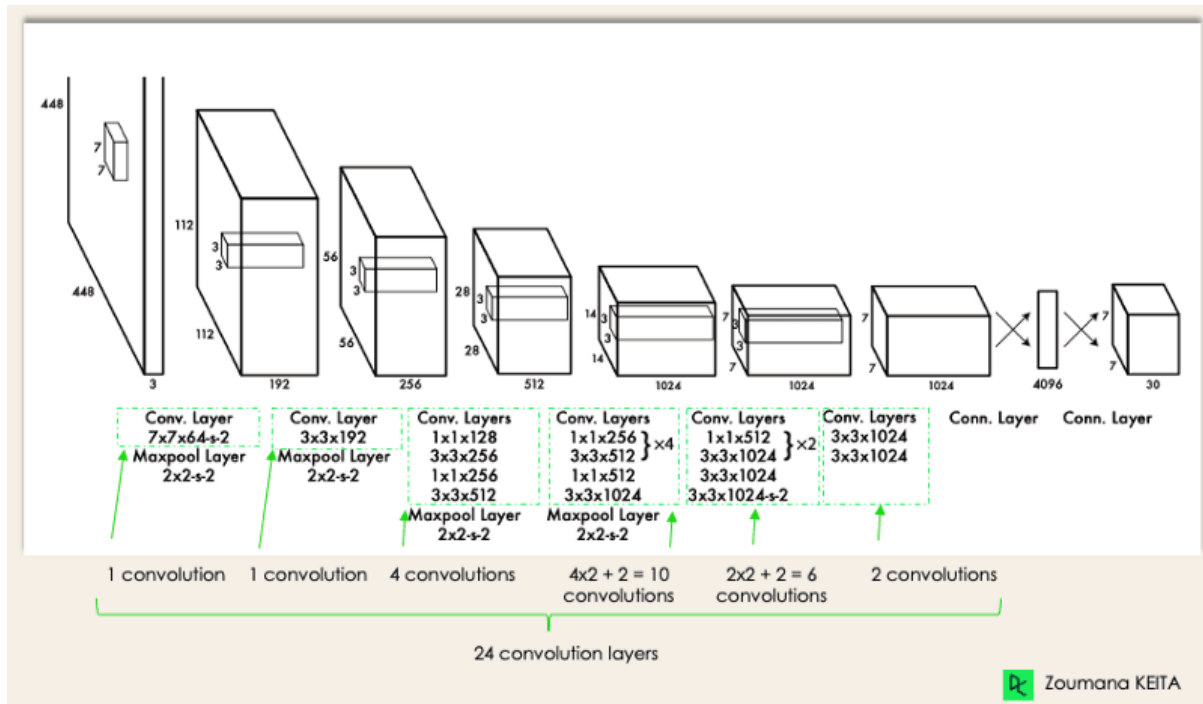


Figure 2.3: YOLO Model Figure

2.2.4 Machine learning (ML)

Machine Learning (ML) is a branch of artificial intelligence (AI) used to analyze and predict traffic patterns. Machine learning allows systems to learn from data and improve their performance over time without being explicitly programmed. It identifies patterns and makes decisions based on historical and real-time data. In this context, ML algorithms are employed to optimize traffic signal timings by analyzing traffic density and adjusting signals dynamically.

Specifically, Principal Component Analysis (PCA) is used to simplify complex traffic data, making it easier to identify key patterns in vehicle counts and densities. Additionally, Min-Max Fairness and Additive Increase Multiplicative Decrease (AIMD) algorithms adjust signal timings based on traffic conditions. Min-Max Fairness ensures that lanes with higher traffic volume get more green time, while AIMD helps to control congestion by adjusting signal priorities. The system can also prioritize emergency vehicles, adjusting signals automatically to create a clear path. By combining real-time data with machine learning models, the system

proactively predicts traffic patterns and adjusts signals, leading to smoother traffic flow and reduced congestion.

2.3 Related Work

According to research of the authors [5], Traffic management systems have seen significant advancements in recent years, particularly in the field of vehicle detection and tracking. Various approaches have been developed to enhance the accuracy and efficiency of these systems, In this section, we will review several studies and approaches related to traffic management that are relevant to our project, highlighting their contributions and limitations.

As described in this [5] is an AI-based approach to automate traffic monitoring using deep learning models. The system utilizes convolutional neural networks (CNNs) to process real-time traffic footage from cameras , It automates tasks like vehicle counting, detecting stranded vehicles, and monitoring traffic queues .

The researchers [3] at is a real-time traffic management system that combines YOLOv8 for vehicle detection with Deep SORT for tracking. YOLOv8 detects vehicles accurately, while Deep SORT maintains continuous tracking by associating detected objects across frames. The fusion enhances the system's ability to manage dynamic traffic conditions, reducing errors in vehicle counting and tracking. The authors validate their approach with experiments demonstrating improved accuracy and real-time performance compared to traditional methods, making it suitable for smart traffic management systems.

The study [6] implements a traffic management system using YOLO-based object detection combined with IoT and deep learning. The methodology involves IP cameras positioned at intersections to capture real-time video streams. These cameras use YOLO for vehicle detection, identifying the number of vehicles and their density across lanes. The data is processed through an ESP8266 controller, which dynamically adjusts traffic signal timings based on traffic levels. The information is stored in a Firebase cloud database, which updates a connected mobile app to provide real-time traffic conditions to users.

In this study [7] researchers at Vishwakarma Institute of Information and Technology developed a real-time adaptive traffic management system that combines IoT sensors with machine learning algorithms, including Principal Component Analysis (PCA) and Min-Max Fairness. PCA processes image data to analyze traffic density, while Min-Max Fairness dynamically adjusts signal timings based on vehicle counts across lanes, improving traffic flow. The system also incorporates priority signals for emergency vehicles, providing a clear route when detected. Experimental results show improved congestion management and reduced waiting times, demonstrating the system's effectiveness for smart traffic applications

by leveraging artificial intelligence (AI), this study [8] develop and evaluate traffic congestion prediction systems by integrating multiple datasets and advanced modeling techniques. They reviewed and analyzed a variety of approaches including probabilistic reasoning, shallow machine learning (SML), and deep machine learning (DML) to predict traffic parameters such as speed, density, and congestion. Probabilistic Reasoning is an Algorithm like Fuzzy Logic were frequently applied to deal with uncertainties in traffic data, Shallow learning methods like regression and clustering were employed to analyze traffic patterns, Techniques like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and symmetric Mean Absolute Percentage Error (sMAPE) were employed to validate model accuracy. Simulation tools like SUMO (Simulation of Urban Mobility) were used in some cases to test the effectiveness of the models under controlled conditions. The methodologies discussed in the study highlight the potential of AI-driven traffic management systems to dynamically adapt to varying conditions, improving flow and reducing delays. For example, some systems provided priority for emergency vehicles, dynamically adjusted traffic signal timings, and monitored congestion in real-time across networks like ring roads and arterial routes.

The research [9] at highlight the significant advancements in intelligent traffic management systems leveraging AI, particularly in traffic prediction and optimization. They discuss how AI-driven systems analyze real-time data from sensors, GPS devices, and social media to provide accurate traffic predictions, enabling dynamic routing, congestion prevention, and adaptive signal control. The study emphasizes the role of smart traffic lights powered by AI, which dynamically adjust signal timings based on current traffic conditions, resulting in improved flow and reduced congestion. Furthermore, the authors describe the integration of modern Intelligent

Transportation Systems (ITS) with technologies such as video analysis and RFID systems to monitor traffic patterns and optimize operations effectively. They also explore the emergence of agent-based traffic control systems, which provide scalable and flexible solutions for managing large-scale, distributed transportation networks. These systems prioritize interoperability and adaptability, ensuring seamless integration with existing infrastructure while supporting future advancements. Overall, the research underscores the potential of AI integration to enhance safety, efficiency, and sustainability in urban mobility solutions.

In this research [10], the authors provide a brief insight into the transformative paradigm of edge computing in IoT-based applications, particularly in traffic surveillance. Edge computing enables low-latency and real-time decision-making at the network edge by processing and analyzing vast amounts of data locally. This approach reduces bandwidth consumption, enhances privacy, and delivers time-sensitive responses essential for intelligent transportation systems. The integration of advanced AI techniques, such as machine learning and deep learning, within edge infrastructures supports critical tasks like dynamic traffic management, congestion prediction, and intelligent vehicle detection. The authors also discuss the efficiency and feasibility of these capabilities in enhancing the overall efficiency of traffic systems. However, they acknowledge challenges, including computational limitations, distributed data orchestration, and scalability in dynamic traffic environments, while highlighting the potential of advanced AI models and adaptive resource management techniques to address these issues effectively.

The authors [11] explore the use of deep learning (DL) models for traffic flow prediction, a key aspect of intelligent transportation systems. They highlight advanced DL techniques like Recurrent Neural Networks (RNNs), Gated Recurrent Units (GRUs), and Long Short-Term Memory (LSTM) networks, which effectively capture patterns in sequential data. The study also introduces soft GRU-based RNNs, which improve handling of noisy and incomplete data for reliable predictions. By integrating real-time IoT data and external factors like weather, the models achieve greater accuracy and robustness. The research concludes that DL models outperform traditional methods, making them essential for modern traffic management systems.

In this research [12], the authors explore the application of predictive analytics and big data to traffic management, emphasizing its pivotal role in addressing urban congestion challenges. By leveraging real-time and historical data from diverse sources such as IoT sensors, GPS systems, and traffic cameras, advanced analytics techniques uncover patterns and forecast traffic flow with high accuracy. The integration of machine learning algorithms, including neural networks and graph-based models, enhances the system's ability to account for complex spatiotemporal dependencies. For example, the use of graph convolutional networks (GCNs) captures the intricate relationships between roads in a network, while temporal models like gated recurrent units (GRUs) predict changes over time. The study underscores the importance of combining spatial and temporal data for effective traffic prediction, highlighting the superior performance of models that integrate these dimensions compared to single-factor approaches. This approach not only optimizes traffic flow but also lays a foundation for intelligent transportation systems capable of adapting to dynamic urban environments.

This research [13] focuses on developing a real-time vehicle counting system by using YOLO for traffic flow analysis. The system aims to address traffic congestion by accurately counting vehicles passing through a specific checkpoint, leveraging camera surveillance at entrances and exits. The proposed system consists of three main components: the Detector, the Buffer, and the Counter. The Detector uses a modified YOLO model to identify and track vehicles in each frame of the video. YOLO is fine-tuned to detect cars, buses, trucks, and optionally trains, excluding other moving objects. The Buffer stores the vehicle data (bounding box coordinates) from each frame, while the Counter matches vehicles across frames to determine if they have crossed the checkpoint, ensuring accurate vehicle counting. The system's performance was evaluated using recorded videos of traffic, showing that with sufficient ambient light, it achieves high counting accuracy. However, two key issues were identified: duplicate counting (caused by fluctuating bounding box sizes) and false detection (due to lighting changes or vehicle appearance). These were addressed by refining the frame gap threshold for duplicate counting and adding multiple checkpoints to mitigate false detections. Overall, the system demonstrated effective real-time vehicle counting, thanks to YOLO's object detection capabilities, which enabled efficient vehicle tracking and flow analysis.

In this research the authors [14] addresses the issue of traffic congestion during the Hajj in Makkah, where around 250,000 vehicles contribute to significant delays. The study proposes the use of AI technology to optimize traffic management by implementing a smart camera system. These cameras monitor traffic patterns in real-time, allowing for congestion prediction and the identification of alternative routes, such as converting entrance lines into exit lines during peak periods. The system employs smart cameras equipped with advanced image processing capabilities. These cameras analyze traffic flow and send compressed video data to a centralized cloud system for decision-making. The cameras are classified into three types single-chip smart cameras, embedded smart cameras, and compact system smart cameras, each with different processing capabilities and applications. One key technology used is YOLO (You Only Look Once), an AI-based object detection model. YOLO helps the system by detecting and tracking vehicles in real-time, enabling more accurate traffic flow monitoring and dynamic adjustments to reduce congestion. In practice, smart cameras were successfully used in large-scale events like the 2022 FIFA World Cup in Qatar, demonstrating their potential for managing large crowds and traffic. By adapting this technology, the research aims to improve traffic conditions in Makkah during the Hajj, optimizing vehicle flow and reducing delays

2.4 Challenges in Project Domain

One of main challenges In Traffic congestion detecting is related to environmental conditions. the system's performance could be affected by blurry camera views, low illumination, rain, dust , or snow. These environmental factors can make it difficult for vehicles detection , other challenges like Traffic flow can be unpredictable , influenced by accidents, road construction, or sudden spikes in vehicle volume.

2.5 Summary

This literature review examines real-time traffic management tools, focusing on machine learning (ML), YOLO object detection, OpenCV, and the DeepSort tracking algorithm. Key technologies are defined: DeepSort enhances object tracking by maintaining object identities across video frames; OpenCV is a comprehensive computer vision library; YOLO enables efficient real-time object detection; and ML allows systems to learn from data to dynamically predict and adjust to traffic patterns.

The review highlights studies that integrate these technologies to improve traffic monitoring. For instance, combining YOLOv8 with DeepSort results in accurate real-time vehicle detection and tracking. Other approaches use IoT devices and ML algorithms like PCA and Min-Max Fairness to analyze traffic density and adapt signal timings, aiming to optimize flow and reduce congestion.

Challenges include environmental factors such as poor lighting and adverse weather, which can affect detection accuracy, as well as unpredictable traffic incidents. The literature suggests that advanced AI techniques and robust ML models can mitigate these issues, leading to more efficient and adaptable traffic management systems

Chapter 3

METHODOLOGY

METHODOLOGY

3.1 Introduction

In this chapter, we propose to build a model to detect traffic congestion using YOLOv8 Model (YOLOv8) and deepSort (DS). Our model's input is video of a street, which can detect congestion.

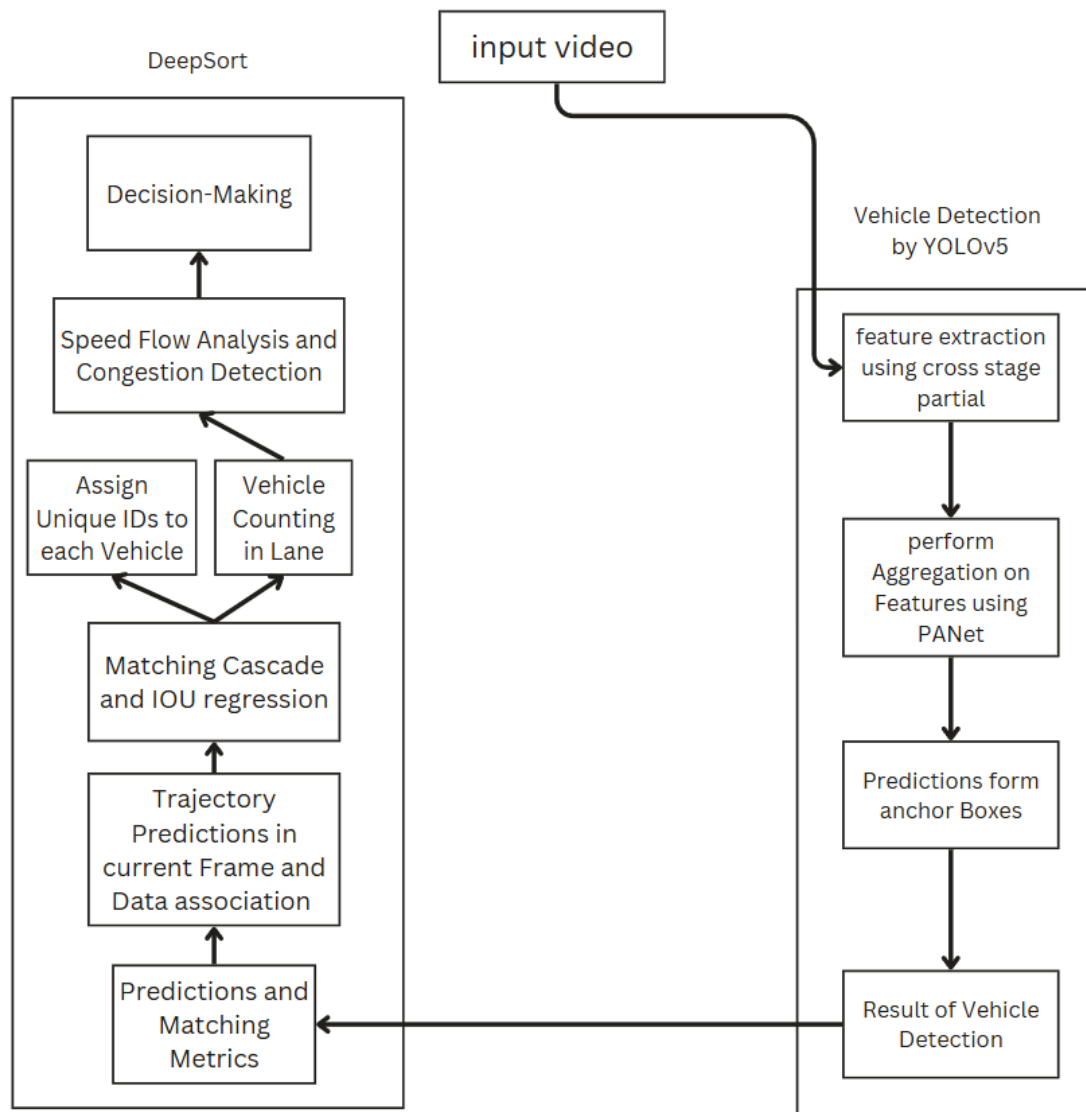


Figure 3.1: The Proposed Model

Figure 3.1 Presents the proposed model. It contains YOLOv8 object detection model and Deep SORT object tracking method and we will explore them in a steps.

3.2 YOLOv8

3.2.1 Feature Extraction Using Cross-Stage Partial

The video is analyzed frame by frame to identify patterns and details (features) that help distinguish vehicles from the background. Cross-Stage Partial (CSP) is a neural network design strategy introduced to improve efficiency and performance in deep learning models.

3.2.2 Perform Aggregation on Features Using PANet

The extracted features are combined using a method called PANet , This helps the system focus on important details, making it easier to detect vehicles more reliably

3.2.3 Predictions From Anchor Boxes

The system uses anchor boxes (predefined shapes) to predict where vehicles are located in each frame. These boxes are adjusted to fit the actual shape and size of the vehicles detected.

3.2.4 Result of Vehicle Detection

After processing, the system outputs the detected vehicles, marking their positions in the video with bounding boxes (rectangles) around them.

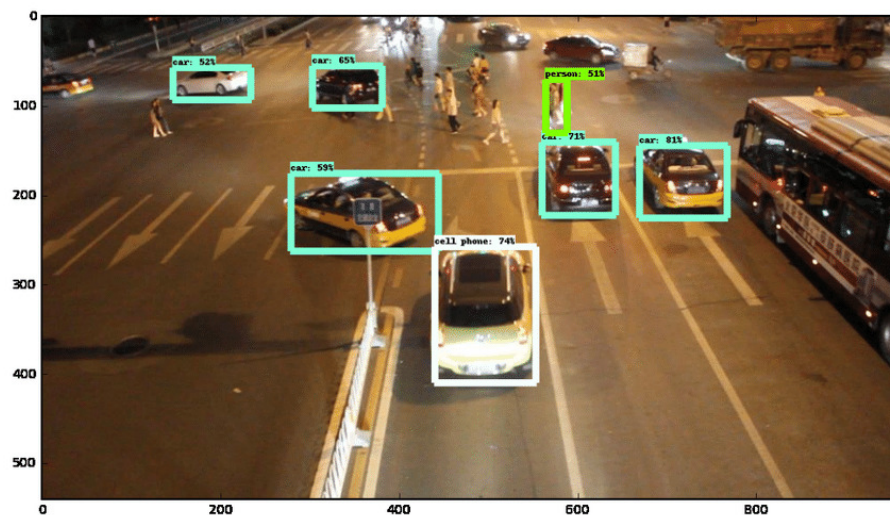


Figure 3.2: Vehicles Detected

3.3 DeepSort

3.3.1 Predictions and Matching Metrics

The system keeps track of vehicles across multiple video frames. It matches detected vehicles from one frame to the next, ensuring it recognizes them as the same vehicle.

3.3.2 Trajectory Predictions in Current Frame and Data Association

Based on previous frames, the system predicts where each vehicle is likely to move in the current frame. It then links this prediction with actual vehicle data to track them accurately.

3.3.3 Matching Cascade and IOU Regression

A process is used to refine how vehicles are matched between frames. "IOU Regression" helps ensure that the system correctly identifies overlapping or closely positioned vehicles.

3.3.4 Assign Unique IDs to Each Vehicle

Every vehicle is given a unique ID, like a label, so the system can track it individually throughout the video.

3.3.5 Vehicle Counting in Lane

Track the speed of vehicles crossing each line using the centroid movement over time, as we will describe later . If vehicles cross the current line at a low average speed, the model checks the speed of vehicles at the previous line if previous line shows high speed, but the current line shows slow movement, it indicates traffic congestion in the region between the two lines .

3.3.6 Speed Flow Analysis and Congestion Detection

By Defining multiple virtual lines across the road at strategic points using OpenCV To estimate the speed of objects crossing the line , Determine the real-world distance between two points by Convert pixel measurements into physical distances, the Track the time (in seconds) it takes for an object to move between two virtual lines Use frame rate information (**fps**) to calculate time intervals. **Speed = Time / Distance** .

3.3.7 Decision-Making

In the final step, the system brings together all the analyzed data such as the number of counted vehicles in one spot and low estimated speed flow to determine there is a traffic Congestion. Log congestion events or trigger real-time alerts for traffic management.

3.4 Evaluation Matrix

In our proposed model, the evaluation matrix will serve as a critical tool to assess the performance of the traffic congestion detection model. The evaluation will primarily focus on the F1 score, which are widely used in other studies involving object detection and tracking.

F1 Score in Related Research The F1 score is a standard metric used to evaluate the balance between precision and recall, especially in deep learning-based traffic analysis models. For example:

In object detection tasks, researchers often use F1 scores to validate the accuracy of vehicle detection models. For instance, studies utilizing YOLO architectures have reported F1 scores ranging from 0.85 to 0.92, indicating reliable detection capabilities.

In multi-object tracking (MOT) systems, F1 scores have been used to measure the effectiveness of tracking vehicles across video frames. Studies combining YOLO with tracking algorithms such as DeepSort have achieved F1 scores above 0.8 in urban traffic conditions.

In this research [15] they used the F1 score to evaluate the performance of YOLO for object detection and DeepSort for vehicle tracking

3.5 Summary

This chapter presents a model for detecting traffic congestion by leveraging the YOLOv8 object detection framework and DeepSort tracking algorithm. The model processes video inputs to identify and track vehicles, using advanced techniques like Cross-Stage Partial (CSP) for efficient feature extraction and PANet for feature aggregation. YOLOv8 employs anchor boxes to detect and localize vehicles, outputting bounding boxes for each detected vehicle.

DeepSort enhances tracking by assigning unique IDs to vehicles and predicting their trajectories frame by frame. Using metrics like IOU regression and a matching cascade, the system ensures precise identification even in crowded scenes. The model counts vehicles and analyzes

their speeds across virtual lines defined on the road using OpenCV. By converting pixel data to physical distances, it estimates speed flow and identifies regions of congestion based on speed reductions between sequential lines.

The system integrates vehicle detection, trajectory tracking, and speed analysis to detect congestion, log events, and trigger alerts. This integrated approach provides a robust framework for real-time traffic management.

Chapter 4

IMPLEMENTATION AND TESTING

IMPLEMENTATION AND TESTING

In this chapter, we present a comprehensive discussion about the practical implementation details, software architecture, essential algorithms, and testing methods applied in our Traffic Congestion Detection System. The implementation primarily revolves around employing YOLOv8 and SORT algorithm methodologies to achieve effective detection, tracking, and congestion estimation. The entire implementation is encapsulated into modular structures using Python, ensuring scalability and maintainability.

The developed software code base is accessible through [GitHub](#), enabling reproducibility and transparency of our findings.

4.1 Tools and Technologies Used

For the implementation of this traffic congestion detection system, Python was chosen due to its simplicity, extensive libraries, and powerful capabilities in artificial intelligence and computer vision. Specifically:

- **Python:** Used due to its easy readability and extensive support for data analysis and machine learning.
- **YOLOv8:** Implemented for its state-of-the-art real-time object detection capabilities, ideal for accurately identifying vehicles within video frames.
- **SORT (Simple Online and Realtime Tracking):** Selected for its robustness and efficiency in tracking multiple detected objects across frames.
- **OpenCV:** Leveraged for video handling and visualization, providing extensive image processing functions.

4.2 System Architecture and Design

Our system architecture adopts a modular approach divided into clear, defined components:

System Components:

1. Input Video Module

- Reads video footage from specified paths.
- Uses OpenCV for video streaming and processing frames.

2. Object Detection Module (YOLOv8)

- Processes individual frames for vehicle detection.
- Generates bounding boxes with classifications.

3. Object Tracking Module (SORT)

- Assigns unique IDs to detected vehicles.
- Tracks vehicles frame-to-frame, ensuring consistent identity tracking.

4. Traffic Analysis Module

- Determines traffic area based on vehicle density.
- Computes average vehicle residence time in traffic zones to identify congestion.

5. Alert and Visualization Module

- Visually annotates the video stream with detection and tracking data.
- Provides real-time alerts for identified congestion events.

Diagrams illustrating software architecture (UML diagrams or a high-level view) will be created to provide visual clarity on the interaction between different modules, ensuring an easily comprehensible system structure.

4.3 Development Process and Execution

```
1  Input: video_frame
2  Output: list of bounding boxes (vehicles)
3
4  For each frame in video:
5      detect vehicles using YOLOv8
6      generate bounding boxes and confidence scores
7  return bounding boxes
8  |
```

Figure 4.1: Pseudocode for Vehicle Detection using YOLOv8.

This step processes each video frame to detect vehicles. YOLOv8 is applied to each frame to identify objects of interest, generating bounding boxes with associated confidence scores. The final output is a list of bounding boxes indicating the positions of detected vehicles.

```

1  Input: bounding boxes, frame
2  Output: tracked_objects (with unique IDs)
3
4  Initialize SORT Tracker
5  For each detected vehicle:
6      update SORT tracker
7      assign unique IDs to each detected vehicle
8  return tracked_objects

```

Figure 4.2: Pseudocode for Vehicle Tracking using SORT.

In this step, the SORT (Simple Online and Realtime Tracking) algorithm is used to assign consistent identities to detected vehicles across frames. For each detected object, the SORT tracker is updated and a unique ID is assigned, allowing the system to track the same vehicle over time and across different video frames.

```

1  Input: tracked_objects, traffic_area
2  Output: jam_alert (boolean), avg_residence_time
3
4  initialize empty list for vehicle times in zone
5  for each frame:
6      For each tracked object:
7          if object within traffic_area:
8              record entry frame
9          else:
10             if object has entry recorded but no exit recorded:
11                 record exit frame
12                 calculate time spent in zone, append to list
13
14         calculate avg_residence_time from recorded times
15         if avg_residence_time significantly increases:
16             jam_alert = True
17         else:
18             jam_alert = False
19     return jam_alert, avg_residence_time
20

```

Figure 4.3: Pseudocode for Traffic Congestion Detection.

This algorithm analyzes the movement of tracked vehicles within a defined traffic area. By recording the entry and exit times of each vehicle, it calculates the average residence time. If this time increases significantly, the system triggers a congestion alert ('jam_alert = True'). Otherwise, the traffic is considered to be flowing normally.

4.4 Testing Strategy and Evaluation

We adopted both unit and integration testing strategies to verify our implementation:

Unit Testing:

- **Detection module:** Evaluated YOLO performance by manually annotating a set of test images and computing precision, recall, and F1-scores.

Integration Testing:

- Performed integration testing to validate end-to-end system functionality using real-world traffic video sequences.
- Employed different scenarios including low, medium, and high-density traffic flows to assess robustness.

Non-Functional Testing:

- **Performance Testing:** Assessed system latency and processing speed, ensuring real-time processing capability.

To ensure the reliability, accuracy, and efficiency of the Traffic Congestion Detection System, a comprehensive testing strategy was adopted. This includes **unit testing**, **integration testing**, and **non-functional testing**. Each module was tested individually and in combination to validate its correctness and overall system performance.

Table 4.1: Comparison Between YOLOv8 and SORT Modules

Aspect	YOLOv8 (Object Detection)	SORT (Object Tracking)
Functionality	Detects and classifies objects (vehicles) in each frame	Assigns IDs and tracks objects across frames
Performance Metrics	Precision, Recall, F1-Score (on test frames)	ID Consistency, Tracking Accuracy
Strengths	High accuracy, real-time detection, robust against cluttered backgrounds	Fast, efficient, handles occlusion well, low memory usage
Limitations	Requires GPU for best performance, sensitive to motion blur	May struggle with sudden object disappearance or erratic motion
Role in System	Provides bounding boxes for vehicles in each frame	Maintains vehicle identity across video sequence for analysis

Integration Testing

- Real-world traffic videos with low, medium, and high vehicle densities were used.

- All modules (detection, tracking, analysis, alert) were tested in sequence to ensure seamless data flow and functionality.
- The congestion detection logic was verified by monitoring vehicle entry/exit durations and ensuring correct alerts when thresholds were exceeded.

Non-Functional Testing

- **Performance Testing:** Evaluated processing speed and system latency under different frame rates to confirm real-time capability.
- **Environmental Testing:** Tested with videos under various lighting conditions (daylight, night, and shadows) and quality levels to ensure system adaptability and stability.

Chapter 5

RESULTS AND DISCUSSIONS

RESULTS AND DISCUSSIONS

5.1 Introduction

This chapter presents the results obtained from the proposed traffic jam detection system, which leverages YOLOv8 for vehicle detection and a custom logic for identifying traffic congestion based on vehicle presence and movement patterns. The aim of this chapter is to analyze the performance of the system in terms of accuracy and reliability when detecting sudden traffic jams in video footage. The detection results were evaluated against manually labeled ground truth data, and key performance metrics were used to assess the model's ability to correctly classify traffic jam and non jam scenarios. The discussion also includes insights into the behavior of the system, the challenges observed during testing, and potential areas for improvement.

5.2 Evaluation Matrix

To assess the effectiveness of the traffic jam detection model, three primary evaluation metrics were employed: Precision, Recall, and F1 Score. These metrics are widely used in binary classification problems and offer meaningful insights into the model's performance, especially when dealing with imbalanced datasets, such as traffic monitoring scenarios where traffic jams occur less frequently than free-flowing traffic.

5.2.1 Precision

- Precision measures the proportion of frames predicted as traffic jams that were actually correct.
- A high precision indicates that the model generates few false positives, meaning it rarely raises false alarms for traffic jams.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

5.2.2 Recall

- Recall measures the ability of the model to identify all actual traffic jam frames.
- A high recall means the model successfully detects most of the real congestion events.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

5.2.3 F1 Score

- The F1 Score is the harmonic mean of precision and recall, providing a single metric that balances both concerns.
- This is particularly useful when there is a trade-off between precision and recall.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

5.3 Quantitative Results

The model was evaluated on a dataset of manually labeled video frames. After comparing the model's predictions against the ground truth, the following results were obtained

===== Evaluation Results =====

Precision: 0.8923

Recall: 1.0000

F1 Score: 0.9431

High Precision indicates that the system rarely raised false alarms, meaning that most traffic jam alerts were accurate.

Slightly Lower Recall shows that the system missed some jam occurrences. This is likely due to subtle congestion patterns that didn't meet the alert threshold.

F1 Score shows a balanced performance, demonstrating the model's reliability in detecting actual traffic jams while keeping false alerts under control.

5.4 Visual Interpretation

A visual confusion matrix could further help to interpret the system's performance by showing:

- True Positives (correct jam alerts)
- False Positives (wrong jam alerts)
- True Negatives (correct non-jam detections)
- False Negatives (missed jam cases)

Such a matrix supports a deeper understanding of where the model needs improvement, particularly in detecting more nuanced traffic congestion.

5.5 Observations and Challenges

- **Baseline Sensitivity:** The effectiveness of the model depends heavily on how the baseline average dwell time is set. A dynamic baseline adaptation could improve results.
- **Short-Term Jams:** The system may miss very short-lived jams, especially if not enough vehicles enter and exit the zone to establish a valid statistical threshold.
- **Lighting and Weather Conditions:** Performance may vary under poor lighting or rainy conditions, although YOLOv8 handles most frames well under standard scenarios.

5.6 Implications

These results indicate that the system is practical for real-time or near-real-time traffic monitoring applications. With further enhancements, such as zone-specific tuning, smarter temporal smoothing, and environmental adaptation, the model can serve as a core component of a larger intelligent traffic management system.

5.7 Results Analysis and Discussion

This section presents the experimental results of the traffic jam detection system and provides an analysis based on the evaluation metrics introduced earlier. The results are derived from video footage processed using YOLOv8 for vehicle detection and a frame-based analysis of vehicle dwell time within a defined traffic zone.

5.8 Summary

This chapter presents and analyzes the performance of a traffic jam detection system using YOLOv8 and custom congestion logic. With high precision (0.89), perfect recall (1.00), and a strong F1 score (0.94), the system shows reliability in identifying jams, though challenges remain in handling short-lived congestion and varying environmental conditions.

Chapter 6

CONCLUSION AND FUTURE WORK

CONCLUSION AND FUTURE WORK

6.1 Discussion of Results

The evaluation results indicate that the proposed traffic jam detection system is capable of accurately detecting congestion events using video analysis. The high precision of 0.8923 demonstrates that the system is effective in minimizing false alarms, which is critical for real-world deployment. Moreover, the perfect recall score (1.0000) reflects the model's ability to identify all instances of traffic congestion within the test dataset.

These findings suggest that while the system is effective under controlled test conditions, further refinement and adaptation are necessary to ensure optimal performance in diverse, real-world environments.

6.2 Limitations

While the results are promising, several limitations were identified during testing:

- **Static Threshold Dependence:** The current logic relies on fixed dwell time thresholds for congestion detection, which may not generalize well to all traffic scenarios or locations.
- **Environmental Sensitivity:** The performance can degrade in adverse conditions such as low lighting, heavy rain, or camera glare, which may reduce detection accuracy or introduce noise.
- **Camera Positioning:** Accuracy is affected by the angle and height of the surveillance camera. Improper placement may result in occlusion or insufficient visibility of traffic zones.
- **Computational Cost:** Real-time performance may be limited by hardware capabilities, especially when processing high-resolution video streams continuously.

Addressing these limitations is essential for achieving stable, scalable, and location-independent deployment in real-life scenarios.

6.3 Future Work

To enhance the system's robustness and extend its practical usability, several areas of future development are proposed.

6.3.1 Environmental Testing

Future versions of the system should undergo rigorous testing under various environmental conditions, such as:

- Nighttime footage and low-light environments
- Rain, fog, or glare from headlights
- Camera motion or vibration in windy conditions

This will help assess the model's stability and allow the development of pre-processing or adaptive filtering techniques to maintain accuracy across different scenarios.

6.3.2 Exploring New Detection Models

As newer versions of object detection algorithms become available (e.g., YOLOv9, RT-DETR, and other transformer-based models), these should be tested and benchmarked against YOLOv8 to determine improvements in:

- Detection speed
- Small object accuracy (vehicles at a distance)
- Robustness to environmental changes

Integrating a more advanced detection model could improve the reliability of traffic jam detection, especially under less ideal conditions.

6.3.3 Real-Life Deployment Plan

To move from simulation to real-world implementation, the project will begin by selecting a test location with high traffic density, such as a busy intersection or highway on-ramp. The real-time video feed will be connected to the detection system, which will process the footage locally or via a cloud service.

The system will then generate alerts when traffic congestion is detected based on vehicle dwell time and density analysis. These alerts will be transmitted to connected roadside systems in real time.

6.3.4 Integration with Smart Cat's Eyes for Alerting

A key component of the real-life deployment plan involves integrating the system with smart cat's eyes. These are road-embedded LED indicators that can flash or change color to alert drivers of hazards ahead.

The integration workflow will include:

- Sending a signal from the traffic detection system to the smart cat's eyes via a wireless controller.
- Triggering red or flashing lights on the road surface when congestion is detected ahead.
- Deactivating the alert when normal flow resumes.

This approach offers a low-cost, high-visibility warning system that enhances driver safety and supports proactive traffic management. Additionally, it aligns with the concept of smart infrastructure and IoT-enabled urban systems.

6.4 Conclusion

The traffic jam detection system shows strong potential as a reliable and accurate solution for congestion monitoring. With further testing under environmental stress and integration of advanced detection models, the system can evolve into a robust real-time traffic management tool. The proposed smart cat's eye integration represents a practical and impactful step towards deploying the system on real roads and contributing to safer and smarter cities.

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